

Assumptions and basis for calculation

1. Quality & standards: As per customer specification
2. Production capacity: 54,000 units per annum
3. Shifts: Single shift with 75 per cent efficiency. Maximum capacity utilisation on single-shift basis for 300 days a year. During first and second years of operation, capacity utilisation is 60 per cent and 80 per cent, respectively. The unit is expected to achieve full capacity utilisation third year onwards.
4. Rate per PCB unit: ₹

tools, knives, blades, scissors, wire strippers, service tools and brushes

7. Machines for radial and axial components cutting/components forming/components bending/components kinking, electronic components counter, selective soldering and rework systems

8. Humidity-controlled electronic components storage cabinets

Manufacturing process. PCB assembly process consists of several manufacturing and quality control steps depending on the quality and specification requirements of the final product. However, the basic assembly procedure is quite simple.

First, electronic components and the PCB are tested for the required specifications. In the assembly line, electronic components such as resistors, capacitors, coils, diodes, transistors, ICs and SCRs are fitted on the PCB at their appropriate location under different stages. Then the PCB is passed over the heat-wave soldering bath for soldering of components to the PCB.

In the next stage, unwanted component leads are removed and components are checked for proper soldering with the help of a magnifying glass fitted with tubelight. Then with the help of appropriate test jigs, the assembled PCB is tested for its specification before despatch to the customer.

Major steps involved in this procedure are detailed below:

1. **Heat pressing and etching.** Press the layout on the board using heat. Additionally, remove unnecessary copper from the board.

2. **PCB drilling and conductor plating.** Drill holes to match legs of electronic components. Then to go for nickel or tin plating on copper around holes so that components can

be soldered.

3. **Soldering and testing.** Pass the PCB through the heat-wave soldering bath for soldering of components to the PCB. In the next stage, remove unwanted lead-ends of components. Check whether the circuit is giving the required voltage and current. Place electronic components on their respective location and solder them. Finally, with the help of appropriate test jigs, test the assembled PCB for its specification.

Pollution control. The Indian government accords utmost importance to environment pollution control. Small-scale entrepreneurs should have an environment-friendly attitude and adopt pollution control measures by process modification and technology substitution. India having acceded to the Montreal Protocol in September 1992, production and use of ozone-depleting substances (ODS) like chlorofluorocarbons, carbon tetrachloride, halons and methyl chloroform need to be phased out immediately with use of alternative chemicals/solvents. Detailed rules for regulation of ODS phase-out were notified under the Environment Protection Act 1986, on 19th July, 2000.

The following steps may help to control pollution in the electronics industry wherever applicable:

1. Fumes and gases released during hand soldering/wave soldering/dip soldering are harmful to people as well as the environment and end products. Alternative technologies may be used to phase out the existing polluting technologies. Numerous new fluxes have been developed containing 2-10 per cent solids as opposed to the traditional 15-35 per cent solids.

2. CFCs, carbon tetrachloride

and methyl chloroform are used to clean PCBs after assembly in order to remove flux residues left after soldering and various kinds of foams for packaging. These could be replaced with alternative solvents. Chlorinated solvents such as trichloroethylene, perchloroethylene and methylene chloride have been used as effective cleaners in the electronics industry for many years. Organic solvents such as ketones and alcohols are effective in removing both solder fluxes and many polar contaminants.

Skillset requirements. You must have good understanding of:

1. Functionalities of electronic components like transistors, capacitors, resistors, diode and ICs
2. Working of the electronic circuit
3. Different PCB base materials
4. Electronic design, components assembly and circuit testing
5. IPC/Mil standards
6. Circuit board design software like CAD, Allegro, Altium and PADS

Business economics

Fixed capital.

Production machinery and equipment cost = A

Cost of office equipment, furniture, working tables, etc = B

Electrification cost (@ 10 per cent of production machinery cost) = 0.1A

Preoperative expenses = C

Total fixed capital = A + B + C + 0.1A = D

Operational cost per month. Minimum land and building requirements are as follows:

1. Land and building builtup area: 150sq.m

2. Stores: 50sq.m

3. Assembly and testing facility: 100sq.m

Rent for the above-mentioned infrastructure/month = R

Human resource requirements are as follows:

1. Servicing engineer cum manager: 1

2. Accountant/cum computer operator: 1

3. Skilled workers: 6

4. Semi-skilled workers: 4

5. Office boy/watchman: 1

Human resource cost per month = H